Tata Kelola dan Manajemen Teknologi Informasi

Manajemen TI Domain DSS

Team Teaching ISM

S1 Sistem Informasi – Fakultas Rekayasa Industri

Outline

1. **DSS01** Managed **Operations**
2. **DSS02** Managed **Service Requests** and **Incidents**
3. **DSS03** Managed **Problems**
4. **DSS04** Managed **Continuity**
5. **DSS05** Managed **Security** Services
6. **DSS06** Managed **Business Process Controls**



Process and **Organizational Structures** **Components** of DSS Domain

DSS01 - Managed Operations

Purpose: Deliver I&T operational product and service outcomes as planned.

Practice	Short Description
DSS01.01 Perform operational procedures.	Maintain and perform operational procedures and operational tasks reliably and consistently.
DSS01.02 Manage outsourced I&T services.	Manage the operation of outsourced I&T services to maintain the protection of enterprise information and reliability of service delivery.
DSS01.03 Monitor I&T infrastructure.	Monitor the I&T infrastructure and related events. Store sufficient chronological information in operations logs to reconstruct and review time sequences of operations and other activities surrounding or supporting operations.
DSS01.04 Manage the environment.	Maintain measures for protection against environmental factors. Install specialized equipment and devices to monitor and control the environment.
DSS01.05 Manage facilities.	Manage facilities, including power and communications equipment, in line with laws and regulations, technical and business requirements, vendor specifications, and health and safety guidelines.

DSS01 - Managed Operations

Purpose: Deliver I&T operational product and service outcomes as planned.

B. Component: Organizational Structures							
Key Management Practice	**Chief Operating Officer**	**Chief Information Officer**	**Chief Technology Officer**	**Head IT Operations**	**Information Security Manager**	**Privacy Officer**
DSS01.01 Perform operational procedures.	R	A	R	R		
DSS01.02 Manage outsourced I&T services.		A	R	R	R	R
DSS01.03 Monitor I&T infrastructure.		R	A	R	R	
DSS01.04 Manage the environment.		R	A	R	R	
DSS01.05 Manage facilities.		R	A	R	R	
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**					
No related guidance for this component						

DSS02 - Managed Service Requests and Incidents

Purpose: Achieve increased productivity and minimize disruptions through quick resolution of user queries and incidents. Assess the impact of changes and deal with service incidents. Resolve user requests and restore service in response to incidents.

Practice	Short Description
DSS02.01 Define classification schemes for incidents and service requests.	Define classification schemes and models for incidents and service requests.
DSS02.02 Record, classify and prioritize requests and incidents	Identify, record and classify service requests and incidents and assign a priority according to business criticality and service agreements.
DSS02.03 Verify, approve and fulfill service requests.	Select the appropriate request procedures and verify that the service requests fulfill defined request criteria. Obtain approval, if required, and fulfill the requests.
DSS02.04 Investigate, diagnose and allocate incidents.	Identify and record incident symptoms, determine possible causes, and allocate for resolution.
DSS02.05 Resolve and recover from incidents.	Document, apply and test the identified solutions or workarounds. Perform recovery actions to restore the I&T-related service.
DSS02.06 Close service requests and incidents.	Verify satisfactory incident resolution and/or fulfilment of requests, and close.
DSS02.07 Track status and produce reports.	Regularly track, analyze and report incidents and fulfilment of requests. Examine trends to provide information for continual improvement.

DSS02 - Managed Service Requests and Incidents

Purpose: Achieve increased productivity and minimize disruptions through quick resolution of user queries and incidents. Assess the impact of changes and deal with service incidents. Resolve user requests and restore service in response to incidents.

B. Component: Organizational Structures							
Key Management Practice	**Chief Technology Officer**	**Business Process Owners**	**Head Development**	**Head IT Operations**	**Service Manager**	**Information Security Manager**
DSS02.01 Define classification schemes for incidents and service requests.	A		R	R	R	
DSS02.02 Record, classify and prioritize requests and incidents.	A			R	R	
DSS02.03 Verify, approve and fulfil service requests.	A	R	R	R	R	
DSS02.04 Investigate, diagnose and allocate incidents.	A	R		R	R	
DSS02.05 Resolve and recover from incidents.	A		R	R	R	R
DSS02.06 Close service requests and incidents.	A			R	R	R
DSS02.07 Track status and produce reports.	A			R	R	
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**					
ISO/IEC 27002:2013/Cor.2:2015(E)	16.1.1 Responsibilities and procedures					

DSS03 - Managed Problems

Purpose: Increase availability, improve service levels, reduce costs, improve customer convenience and satisfaction by reducing the number of operational problems, and identify root causes as part of problem resolution.

Practice	Short Description
DSS03.01 Identify and classify problems.	Define and implement criteria and procedures to identify and report problems. Include problem classification, categorization and prioritization.
DSS03.02 Investigate and diagnose problems.	Investigate and diagnose problems using relevant subject matter experts to assess and analyze root causes.
DSS03.03 Raise known errors.	As soon as root causes of problems are identified, create known-error records, document appropriate workarounds and identify potential solutions.
DSS03.04 Resolve and close problems.	Identify and initiate sustainable solutions addressing the root cause. Raise change requests via the established change management process, if required, to resolve errors. Ensure that the personnel affected are aware of the actions taken and the plans developed to prevent future incidents from occurring.
DSS03.05 Perform proactive problem management.	Collect and analyze operational data (especially incident and change records) to identify emerging trends that may indicate problems. Log problem records to enable assessment.

DSS03 - Managed Problems

Purpose: Increase availability, improve service levels, reduce costs, improve customer convenience and satisfaction by reducing the number of operational problems, and identify root causes as part of problem resolution.

B. Component: Organizational Structures								
Key Management Practice	Executive Committee	Chief Information Officer	Chief Technology Officer	Head Development	Head IT Operations	Service Manager	Information Security Manager
DSS03.01 Identify and classify problems.		R	A	R	R	R	
DSS03.02 Investigate and diagnose problems.			A		R	R	R
DSS03.03 Raise known errors.			A		R	R	R
DSS03.04 Resolve and close problems.			A		R	R	
DSS03.05 Perform proactive problem management.	R		A		R	R	
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**						
No related guidance for this component							

DSS04 - Managed Continuity

Purpose: Adapt rapidly, continue business operations and maintain availability of resources and information at a level acceptable to the enterprise in the event of a significant disruption (e.g., threats, opportunities, demands)

Practice	Short Description
DSS04.01 Define the business continuity policy, objectives and scope.	Define business continuity policy and scope, aligned with enterprise and stakeholder objectives, to improve business resilience.
DSS04.02 Maintain business resilience.	Evaluate business resilience options and choose a cost-effective and viable strategy that will ensure enterprise continuity, disaster recovery and incident response in the face of a disaster or other major incident or disruption.
DSS04.03 Develop and implement a business continuity response.	Develop a business continuity plan (BCP) and disaster recovery plan (DRP) based on the strategy. Document all procedures necessary for the enterprise to continue critical activities in the event of an incident.
DSS04.04 Exercise, test and review the business continuity plan (BCP) and disaster response plan (DRP).	Test continuity on a regular basis to exercise plans against predetermined outcomes, uphold business resilience and allow innovative solutions to be developed.
DSS04.05 Review, maintain and improve the continuity plans.	Conduct a management review of the continuity capability at regular intervals to ensure its continued suitability, adequacy and effectiveness. Manage changes to the plans in accordance with the change control process to ensure that continuity plans are kept up to date and continually reflect actual business requirements.
DSS04.06 Conduct continuity plan training.	Provide all concerned internal and external parties with regular training sessions regarding procedures and their roles and responsibilities in case of disruption
DSS04.07 Manage backup arrangements.	Maintain availability of business-critical information.
DSS04.08 Conduct post-resumption review.	Assess the adequacy of the business continuity plan (BCP) and disaster response plan (DRP) following successful resumption of business processes and services after a disruption.

DSS04 - Managed Continuity

Purpose: Adapt rapidly, continue business operations and maintain availability of resources and information at a level acceptable to the enterprise in the event of a significant disruption (e.g., threats, opportunities, demands)

B. Component: Organizational Structures														
Key Management Practice	Executive Committee	Chief Operating Officer	Chief Information Officer	Chief Technology Officer	Chief Information Security Officer	Business Process Owners	Data Management Function	Head Architect	Head Development	Head IT Operations	Service Manager	Information Security Manager	Business Continuity Manager
DSS04.01 Define the business continuity policy, objectives and scope.	R	A	R		R	R				R	R		R
DSS04.02 Maintain business resilience.	R	A	R			R		R		R		R	R
DSS04.03 Develop and implement a business continuity response.			R	R		R				R		R	A
DSS04.04 Exercise, test and review the business continuity plan (BCP) and disaster response plan (DRP).			R	R		R				R		R	A
DSS04.05 Review, maintain and improve the continuity plans.		A	R	R	R	R				R			R
DSS04.06 Conduct continuity plan training.			R	R		R			R	R		R	A
DSS04.07 Manage backup arrangements.				A			R			R		R	R
DSS04.08 Conduct post-resumption review.			R	R	R	R				R			A
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**												
No related guidance for this component													

DSS05 - Managed Security Services

Purpose: **Minimize the business impact of operational information security vulnerabilities and incidents.**

Practice	Short Description
DSS05.01 Protect against malicious software.	Implement and maintain preventive, detective and corrective measures (especially up-to-date security patches and virus control) across the enterprise to protect information systems and technology from malicious software (e.g., ransomware, malware, viruses, worms, spyware, spam).
DSS05.02 Manage network and connectivity security.	Use security measures and related management procedures to protect information over all methods of connectivity.
DSS05.03 Manage endpoint security.	Ensure that endpoints (e.g., laptop, desktop, server, and other mobile and network devices or software) are secured at a level that is equal to or greater than the defined security requirements for the information processed, stored or transmitted.
DSS05.04 Manage user identity and logical access.	Ensure that all users have information access rights in accordance with business requirements. Coordinate with business units that manage their own access rights within business processes.
DSS05.05 Manage physical access to I&T assets.	Define and implement procedures (including emergency procedures) to grant, limit and revoke access to premises, buildings and areas, according to business need. Access to premises, buildings and areas should be justified, authorized, logged and monitored. This requirement applies to all persons entering the premises, including staff, temporary staff, clients, vendors, visitors or any other third party.
DSS05.06 Manage sensitive documents and output devices.	Establish appropriate physical safeguards, accounting practices and inventory management regarding sensitive I&T assets, such as special forms, negotiable instruments, special-purpose printers or security tokens.
DSS05.07 Manage vulnerabilities and monitor the infrastructure for security-related events.	Using a portfolio of tools and technologies (e.g., intrusion detection tools), manage vulnerabilities and monitor the infrastructure for unauthorized access. Ensure that security tools, technologies and detection are integrated with general event monitoring and incident management.

DSS05 - Managed Security Services

Purpose: Minimize the business impact of operational information security vulnerabilities and incidents.

B. Component: Organizational Structures								
Key Management Practice	**Chief Information Officer**	**Chief Information Security Officer**	**Business Process Owners**	**Head Human Resources**	**Head Development**	**Head IT Operations**	**Information Security Manager**	**Privacy Officer**
DSS05.01 Protect against malicious software.		A	R	R	R	R	R	
DSS05.02 Manage network and connectivity security.		A			R	R	R	
DSS05.03 Manage endpoint security.		A			R	R	R	
DSS05.04 Manage user identity and logical access.		A	R			R	R	R
DSS05.05 Manage physical access to I&T assets.		A				R	R	R
DSS05.06 Manage sensitive documents and output devices.	A					R		R
DSS05.07 Manage vulnerabilities and monitor the infrastructure for security-related events.		A				R	R	R

Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**
No related guidance for this component	

DSS06 - Managed Business Process Controls

Purpose: Maintain information integrity and the security of information assets handled within business processes in the enterprise or its outsourced operation.

Practice	Short Description
DSS06.01 Align control activities embedded in business processes with enterprise objectives.	Continually assess and monitor the execution of business process activities and related controls (based on enterprise risk), to ensure that processing controls align with business needs.
DSS06.02 Control the processing of information.	Operate the execution of the business process activities and related controls, based on enterprise risk. Ensure that information processing is valid, complete, accurate, timely and secure (i.e., reflects legitimate and authorized business use).
DSS06.03 Manage roles, responsibilities, access privileges and levels of authority.	Manage business roles, responsibilities, levels of authority and segregation of duties needed to support the business process objectives. Authorize access to all information assets related to business information processes, including those under the custody of the business, IT and third parties. This ensures that the business knows where the data are and who is handling data on its behalf.
DSS06.04 Manage errors and exceptions.	Manage business process exceptions and errors and facilitate remediation, executing defined corrective actions and escalating as necessary. This treatment of exceptions and errors provides assurance of the accuracy and integrity of the business information process.
DSS06.05 Ensure traceability and accountability for information events.	Ensure that business information can be traced to an originating business event and associated with accountable parties. This discoverability provides assurance that business information is reliable and has been processed in accordance with defined objectives.
DSS06.06 Secure information assets.	Secure information assets accessible by the business through approved methods, including information in electronic form (e.g., portable media devices, user applications and storage devices, or other methods that create new assets in any form), information in physical form (e.g., source documents or output reports) and information during transit. This benefits the business by providing end-to-end safeguarding of information.

DSS06 - Managed Business Process Controls

Purpose: Maintain information integrity and the security of information assets handled within business processes in the enterprise or its outsourced operation.

B. Component: Organizational Structures										
Key Management Practice	**Executive Committee**	**Chief Information Officer**	**I&T Governance Board**	**Chief Information Security Officer**	**Business Process Owners**	**Data Management Function**	**Service Manager**	**Information Security Manager**	**Legal Counsel**
DSS06.01 Align control activities embedded in business processes with enterprise objectives.	R		A		R				
DSS06.02 Control the processing of information.		R	A	R	R	R			R
DSS06.03 Manage roles, responsibilities, access privileges and levels of authority.		R	A	R	R			R	
DSS06.04 Manage errors and exceptions.		R		R	A		R		
DSS06.05 Ensure traceability and accountability for information events.		R		R	A				
DSS06.06 Secure information assets.		R		R	A				
Related Guidance (Standards, Frameworks, Compliance Requirements)	**Detailed Reference**								
No related guidance for this component									

Information Input Example of DSS Domain

Information Input Example of DSS Domain

Objective	Practice	Input	Description
DSS01 - Managed Operations	DSS01.01 Perform operational procedures.	BAI05.05	Operation and use plan
DSS02 - Managed Service Requests and Incidents	DSS02.01 Define classification schemes for incidents and service requests.	APO09.03	SLAs
DSS03 - Managed Problems	DSS03.01 Identify and classify problems.	DSS02.04	Problem log
DSS04 - Managed Continuity	DSS04.06 Conduct continuity plan training.	HR	**List of personnel requiring training**
DSS05 - Managed Security Services	DSS05.02 Manage network and connectivity security.	APO01.07	Data classification guidelines
DSS06 - Managed Business Process Controls	DSS06.01 Align control activities embedded in business processes with enterprise objectives.	APO01.07	Data classification guidelines

Thank You

That’s all